IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

ADVANCED ELECTRONIC DEVICES

Tuesday, 29 April 2025, 10:00 AM

Time allowed: 3:00 hours

There are FIVE questions on this paper.

Answer Question ONE and THREE other questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : K. Fobelets
Second Marker(s) : Z. Durrani

© Imperial College London

Constants

permittivity of free space:	$\varepsilon_o = 8.85 \times 10^{-12}$ F/m
permeability of free space:	$\mu_o = 4\pi \times 10^{-7}$ H/m
charge of an electron:	$e = 1.6 \times 10^{-19}$ C (1 eV)
thermal voltage:	$kT/e = 0.026$V at $T = 300$K
intrinsic carrier concentration in Si:	$n_i = 1.45 \times 10^{10}$ cm^{-3} at $T = 300$K
intrinsic carrier concentration in Ge:	$n_i = 2.4 \times 10^{13}$ cm^{-3} at $T = 300$K
dielectric constant of Si:	$\varepsilon_r = 12$
dielectric constant of Ge:	$\varepsilon_r = 16$
dielectric constant of SiO_2:	$\varepsilon_{ox} = 4$
electron affinity of Si:	$\chi = 4.05$ eV
electron affinity of Ge:	$\chi = 4.00$ eV
effective density of states of Si:	$N_C = 2.8 \times 10^{19}$ cm^{-3}
	$N_V = 1.04 \times 10^{19}$ cm^{-3}
effective density of states of Ge:	$N_C = 1.04 \times 10^{19}$ cm^{-3}
	$N_V = 6.0 \times 10^{18}$ cm^{-3}
lattice constant of Si:	a = 0.543 nm
lattice constant of Ge:	a = 0.565 nm
lattice constant of SiC:	a = 0.307 nm
bandgap of Si:	$E_G = 1.12$ eV
bandgap of Ge:	$E_G = 0.66$ eV
thermal conductivity Si:	$\kappa = 1.5$ W/cm C
thermal conductivity Si:	$\kappa = 0.6$ W/cm C

Workfunction of metals

metal	ϕ (V)
Al	4.1
Ni	5.0
Ti	4.3
W	4.6

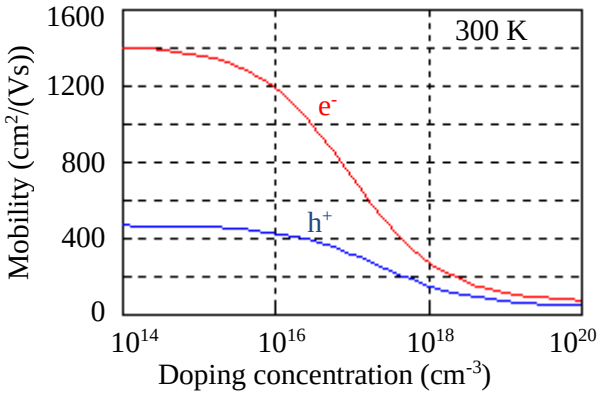


Figure 0: Electron and hole mobility in Si as a function of doping concentration.

Formulae

$p = h/\lambda = \hbar k$ — De Broglie equations

$E = h\nu = \hbar\omega$

$-\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + V\psi = i\hbar\frac{\partial\psi}{\partial t}$ — Schrödinger equation in one-dimension

$p = N_v e^{(E_v - E_F)/kT}$ — Carrier density – Fermi-Dirac statistics

$n = N_c e^{(E_F - E_c)/kT}$

$J_n(x) = e\mu_n n(x)E(x) + eD_n\frac{dn(x)}{dx}$ — Drift – diffusion current density equations for electrons, J_n and holes, J_p

$J_p(x) = e\mu_p p(x)E(x) - eD_p\frac{dp(x)}{dx}$

$I_{DS} = \frac{\mu C_{ox} W}{L}\left((V_{GS} - V_{th})V_{DS} - \frac{V_{DS}^2}{2}\right)$ — Strong inversion MOSFET current

$V_{th} = \phi_m - \phi_s + 2\times\phi_F + \gamma\times\sqrt{2\times\phi_F}$ — Threshold voltage

$\phi_F = \frac{kT}{e}\ln\left(\frac{N_A}{n_i}\right)$ — Fermi potential (difference between intrinsic and Fermi level)

$\gamma = \frac{\sqrt{2e\varepsilon_s N_A}}{C_{ox}}$ — Body effect coefficient

$J = \frac{eD_n n_p}{L_n}\left(e^{\frac{eV}{kT}} - 1\right) + \frac{eD_p p_n}{L_p}\left(e^{\frac{eV}{kT}} - 1\right)$ — Diode diffusion current density

$V_{bi} = \frac{kT}{e}\ln\left(\frac{N_A N_D}{n_i^2}\right)$ — Built-in voltage pn diode

$W_{depl}(V) = \left[\frac{2\varepsilon(V_{bi} - V)}{e}\left(\frac{1}{N_A} + \frac{1}{N_D}\right)\right]^{1/2}$ — Depletion width in pn diode

$S = \frac{dV_{GS}}{dLog(I_{DS})}$ — Sub-threshold swing

$c = c_0\exp\left(\frac{eV}{kT}\right)$ with $\begin{cases} c = p_n \text{ or } n_p \\ c_0 \text{ bulk minority carrier concentration} \end{cases}$ — Minority carrier injection under bias V

$\delta c = \Delta c\exp\left(\frac{-x}{L}\right)$ with $\begin{cases} \delta c = \delta p_n \text{ or } \delta n_p \\ \Delta c \text{ the excess carrier concentration} \\ \text{at the edge of the depletion region} \end{cases}$ — Excess carrier concentration as a function of distance

$L = \sqrt{D\tau}$ — Diffusion length

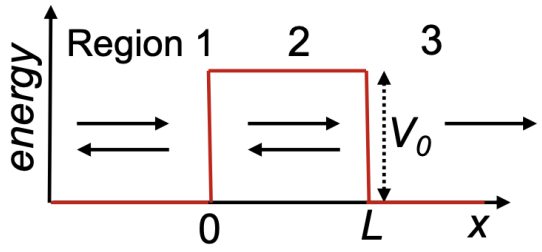
$D = \frac{kT}{e}\mu$ — Einstein relation

$J = \sigma\frac{dV}{dx} + eD\frac{dn}{dx} - \sigma S\frac{dT}{dx}$ — 1D current density with thermal gradient

$J_Q = \frac{J^2}{\sigma} + STJ - \kappa\frac{dT}{dx}$ — 1D heat flux

1. Compulsory question. These are multiple choice questions. Some questions can have multiple answers.

a) Consider an electron incident from the left and tunnelling through a rectangular potential barrier V_0, as shown in the figure below. The directions of the various electron waves are indicated by arrows. [3]



State if the following statements are true or false.

(i) The transmission probability oscillates for electron energy $E < V_0$.

(ii) Incident electrons cannot transmit across the barrier at very low temperature, tending to absolute zero.

(iii) The transmission probability for electron tunnelling depends on both L and V_0.

b) In an Esaki tunnel diode, the current-voltage (I-V) characteristics show a negative differential resistance region for which of the following options? [1]

(i) Reverse bias, applied voltage is small.

(ii) Reverse bias, applied voltage is large.

(iii) Forward bias, applied voltage is small.

(iv) Forward bias, applied voltage is large.

c) What is the main cause of short channel effects? [1]

(i) Mobility degradation due to surface roughness effects.

(ii) The doping in the substrate.

(iii) The depletion width of the drain extending under the gate.

(iv) The small oxide thickness.

d) How can charge sharing be reduced in a short channel MOSFET? [1]

(i) Increasing the doping density in the substrate.

(ii) Increasing the doping density in the source and drain.

(iii) Increasing the doping density in the polySi gate.

(iv) Increasing the oxide doping.

e) Which of the following statements is true in partially-depleted SOI? [2]

(i) All regions of the channel are controlled by the gate.

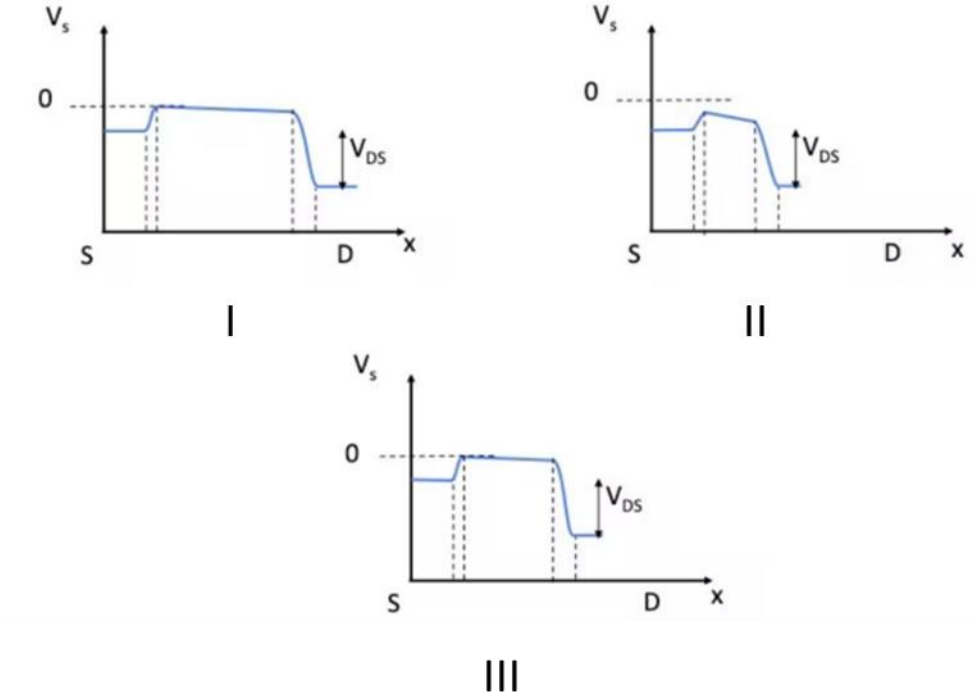
(ii) A region of the channel remains floating for all V_{GS} values.

(iii) All regions of the channel are controlled by the drain.

(iv) The Si body layer is depleted for $V_{GS} = 0$ V.

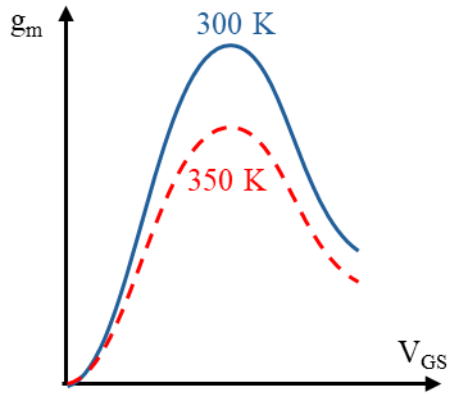
TURN OVER

f) Which of the following figures of the surface potential of the MOSFET from source to drain illustrates DIBL? [2]



(i) I.
(ii) II.
(iii) III.
(iv) None of the three.

g) The figure below gives the tansconductance, g_m as a function of the gate voltage referred to the grounded source, V_{GS} of a MOSFET in the triode region. [2]



Why does g_m decrease for higher temperatures?
(i) Because of the non-ideality of the drift current.
(ii) Because of interface roughness scattering at the SiO_2 interface.
(iii) Because of increased phonon scattering.
(iv) Because the depletion region from the drain influences the source-channel barrier.

TURN OVER

h) Select two statements that are true. (selecting more than two answers will result in a zero mark for the question). [2]
   (i) GAA-FETs are commercially available.
   (ii) GAA-FETs are intended for the sub-7 nm node.
   (iii) GAA-FETs have better DIBL than finFETs.
   (iv) Vertical GAA FETs are faster than horizontal ones.
   (v) GAA-FETS are made by local oxidation processes.

i) Give three methods to increase the current in n-type nanosheet GAA FETs. (selecting more than three answers will result in a zero mark for the question). [3]
   (i) Making the sheets wider.
   (ii) Making the sheets thicker (height of the sheet).
   (iii) Stacking more sheets vertically.
   (iv) Making the gate length longer.
   (v) Applying local tensile strain.
   (vi) Increasing the doping in the nanosheet.

j) What is the difference between global and local strain? [2]
   (i) Local strain: only 1 MOSFET is strained; global strain: all MOSFETs are strained.
   (ii) Local strain: the MOSFET is surrounded by a source and drain material with a different lattice constant; global strain: the channel is strained due to the substrate.
   (iii) Global strain is where only each MOSFET is strained by introducing a material with a different lattice constant in the source and drain whilst in local strain the channel is strained due to the substrate.
   (iv) Local strain is strain applied in finFETs while global strain is applied to bulk MOSFETs.

k) Give three results of implementing strain. (selecting more than three answers will result in a zero mark for the question). [3]
   (i) An increased oxide trap density.
   (ii) A change in the effective mass of the charge carrier in the channel.
   (iii) A change of the mobility of the charge carrier in the channel.
   (iv) A change in the shape of the energy band diagram of the strained layer.
   (v) An increase in oxide capacitance.
   (vi) A decrease in substrate leakage currents.

l) How can you add tools in the workbench of Sentaurus? [1]
   (i) By double clicking on the “no tools” field and selecting the tools.
   (ii) By right clicking on the “no tools” field, selecting “add”.
   (iii) By clicking on the “Edit” button and select “add tools”.

m) What is the Sentaurus command to etch with the same etching rate in all directions? [1]
   (i) Anisotropically.
   (ii) Isotropically.
   (iii) DirectionUniform.

n) What does the DoZero command do in the SDevice script? [1]
   (i) Forcing the voltage to become zero.
   (ii) Forcing the calculation of the voltage at zero time.
   (iii) Force the calculation at the initial voltage on the electrode.

2. The material cross section of a simple long channel nMOS is given in fig. 2.1.

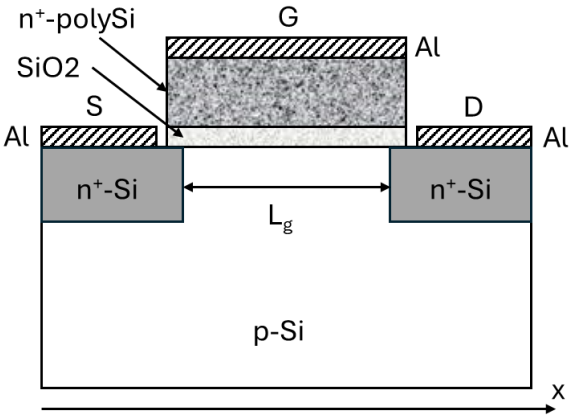


Figure 2.1: Material cross section of a long channel nMOS. L_g is the effective gate length. The gate dielectric is SiO2 and the gate contact is polySi with an Al on top.

a) Sketch the variation of the electrostatic potential $V_x(x)$ from the source junction to the drain junction through an inverted channel in the following two situations. Sketch both curves in the same figure.
   i) For the MOSFET in the triode region (linear region). [2]
   ii) For the MOSFET in saturation at a drain voltage larger than the pinch-off voltage. [2]

b) Draw the energy band diagram, including all relevant parameters from the gate into the bulk for $V_{GS} = V_{DS} = 0$ V with the workfunction of the metal $\phi_m > \phi_{pSi}$ the workfunction of the bulk and a positive charge Q_{it} at the surface of the oxide. [8]

c) Fig. 2.2 gives the transfer characteristics of the long channel MOSFET. Answer the following questions related to fig. 2.2.

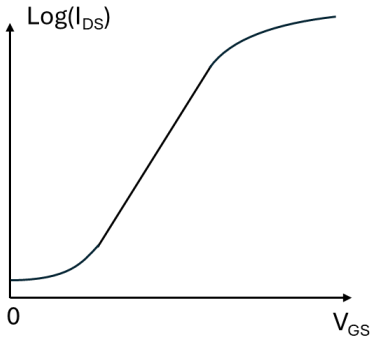


Figure 2.2: Transfer characteristic of the long channel nMOS.

   i) Is this an enhancement or depletion mode MOSFET? [2]
   ii) Which region is the weak inversion region? [2]
   iii) What type of current flows through the channel in weak inversion, electron or hole drift or diffusion? [2]

TURN OVER

d) When downscaling the nMOS given in fig. 2.1 the threshold voltage will change. By what value will the threshold voltage change (ΔV_{th}) if the gate length is scaled down to 14 nm. The gate width is 20 nm, the junction depths t_j = 40 nm and the oxide thickness is 10 nm. The depletion region extending from the source into the channel is 2 nm and that from the drain is 8 nm. The substrate doping is 10^{17} cm^{-3}. You can assume that all doping implantations are rectangular rather than circular. [7]

3. The material cross section of an UTBB-SOI MOSFET is given in fig. 3.1. The metal gate is Ni, the elevated source and drain are SiGe with a Ge concentration of 30%. The substrate doping is $N_A = 5\times10^{16}$ cm^{-3}. The thickness of the undoped Si body is 10 nm while the gate oxide and the BOX are both 5 nm.

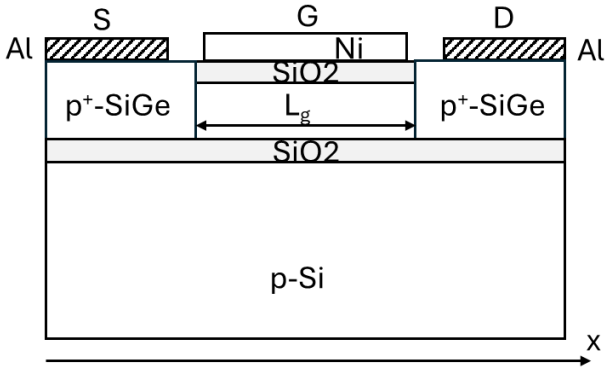


Figure 3.1: The material cross section of an UTBB-SOI MOSFET. The gate contact is Ni. The gate insulator and BOX are SiO2. The source and drain contact are made of epitaxially grown SiGe with an Al metal contact on top. L_g is the gate length.

a) Draw the energy band diagram of the MOSFET in fig. 3.1 from the gate through the body and BOX into the bulk. Include all relevant parameters. The body is sufficiently thick to avoid volume inversion effects. Ensure the relative variations of the potential energies are consistent with the material parameters given. Add the direction of the electric field everywhere where it is different from zero. [10]

b)

i) With respect to your answer of a) is there a channel present at $V_{GS} = 0$ V? Give a brief explanation for your answer. [4]

ii) Is this a fully or partially depleted implementation? [2]

iii) What kind of strain occurs in the channel of the MOSFET? Tensile/compressive/global and/or local strain? [2]

iv) What is the value of the strain assuming the strain in the channel is the same as that close to the source and drain junctions? [2]

v) What impact does the strain have on the mobility of the carriers in the channel? [2]

c) What is the value of the gate voltage that needs to be applied for flat band condition to occur when this voltage is applied both to the gate as well as to the bulk? [3]

4. Figure 4.1 gives a sketch of the 3D shape of the fin of a finFET. The source and drain contact are not drawn but their position relative to the fin is indicated with arrows. The gate length is L_g. The gate insulator and contact material are not drawn and can be assumed to be wrapped around the fin in their usual way.

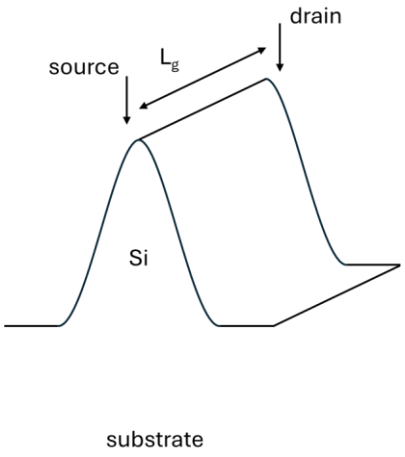


Figure 4.1: a sketch of the geometry of the fin of an n-channel finFET. Gate oxide, gate, source and drain are not drawn.

a) Taking the shape of the fin into account, is this a finFET on bulk or SOI? Give a brief explanation. [4]

b) How many inversion layers are generated in this finFET when a gate voltage larger than the threshold voltage is applied? [2]

c) Answer the following three questions for the case where the fin in Fig. 4.1 is split into a parallel connection of three nanowires, stacked one on top of the other.
   i) Sketch how the nanowire stack might look. Ensure the geometry is derived from that given in fig. 4.1. [4]
   ii) Compare the density of carriers that can be expected in each nanowire at the same gate voltage applied to all three. [3]
   iii) Briefly discuss self-heating in this stack. Where do you expect the temperature to be highest and why? Take both the size and the position of the nanowire within the stack into account. [4]

d) Fig. 4.2 gives a simplified cross section of one of the nanowires. The source is grounded, $V_{GS} > V_{th}$ and $V_{DS} > 0$ V. The relationship between the temperatures is: $T_c < T_M < T_H$.

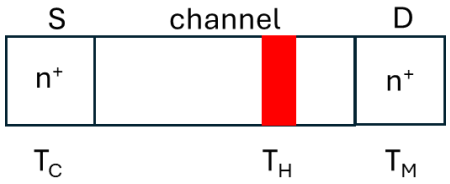


Figure 4.2: A sketch of cross section in the longitudinal direction of one of the nanowire channels between source and drain. The temperature is given in source (T_C) and drain (T_M) area and in the hotspot (T_H).

TURN OVER

i) Redraw fig. 4.2 and sketch the direction of the electron flux due to the electric field. [2]

ii) Redraw fig. 4.2 and sketch the direction of the electron flux due to the temperature gradient. If appropriate give some indication of relative flux density. [4]

iii) Is the saturation current in the nanowire increasing or decreasing due to self-heating? [2]

5. Consider a one-dimensional potential well where the potential energy $V(x)$ is given by:

$$V(x) = 0 \text{ eV} \quad \text{for } 0 \le x \le L$$
$$V(x) = \infty \text{ eV} \quad \text{for } x < 0 \text{ and } x > L$$

a) Solve the one-dimensional, time independent Schrödinger Equation, $-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} - E\psi(x) = 0$, for a particle in the potential well to derive expressions for the particle energy levels E_n in the well. [10]

b) Derive expressions for the corresponding normalised wave functions ψ_n, and the probability density functions corresponding to ψ_n. [8]

c) For a potential well of finite depth, where $V(x) = -V_0$, for $0 \le x \le L$, and $V(x) = 0$ elsewhere, sketch only the wave functions, and the corresponding probability density functions, for n = 1 and 2. How does the peak value of the probability density for n = 1 in this case compare to the peak value in a well with sides of infinite energy? [7]